

A digital illustration of a server room. In the background, there are rows of server racks with glowing blue lights. In the foreground, several translucent, futuristic screens float in the air. These screens display various data visualizations: line graphs, bar charts, network diagrams, and circular progress indicators. The overall color scheme is dark blue and teal, with orange accents on the server racks.

# Setting up Virtualization Software

From installation to troubleshooting—a structured introduction to virtual machines, configuration, and practical troubleshooting.

# What is Virtualization?

## The Basic Principle

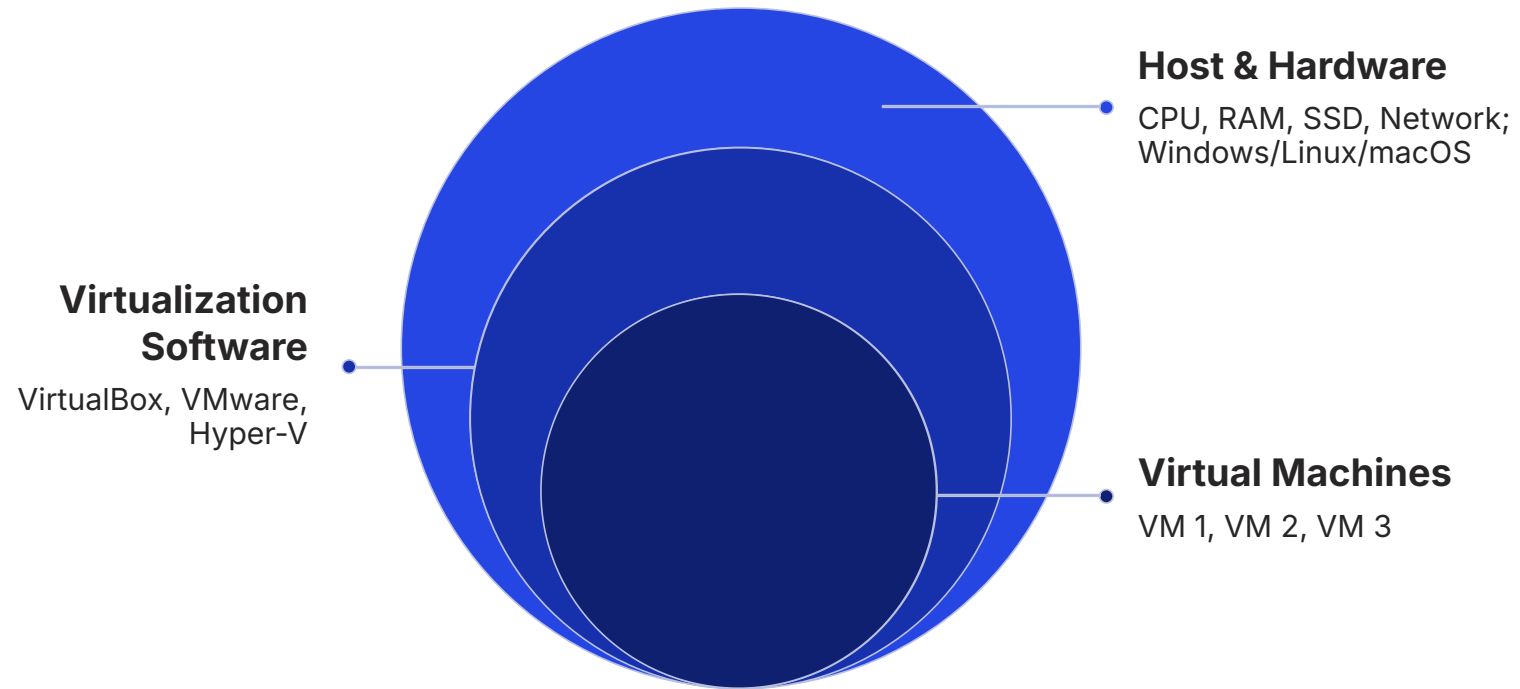
On your actual computer – the **host system** – one or more **virtual machines (VMs)** run. Each VM behaves like its own PC: its own operating system, its own programs, its own network settings.

The virtualization software sits **between the actual hardware and the VM** and distributes resources such as RAM, CPU, storage, and network.

## What is virtualization useful for?

- Testing other operating systems
- Learning in a clean and isolated environment
- Trying out software safely
- Replicating servers, networks, and development environments

# Virtualization Layers



Virtualization software mediates between real hardware and virtual machines – it is the heart of the entire system.

# Which virtualization software is available?

## Oracle VM VirtualBox

Windows, Linux, macOS – very common, great for learning, simple graphical interface.  
**Recommendation for beginners.**

## VMware Workstation Player/Pro

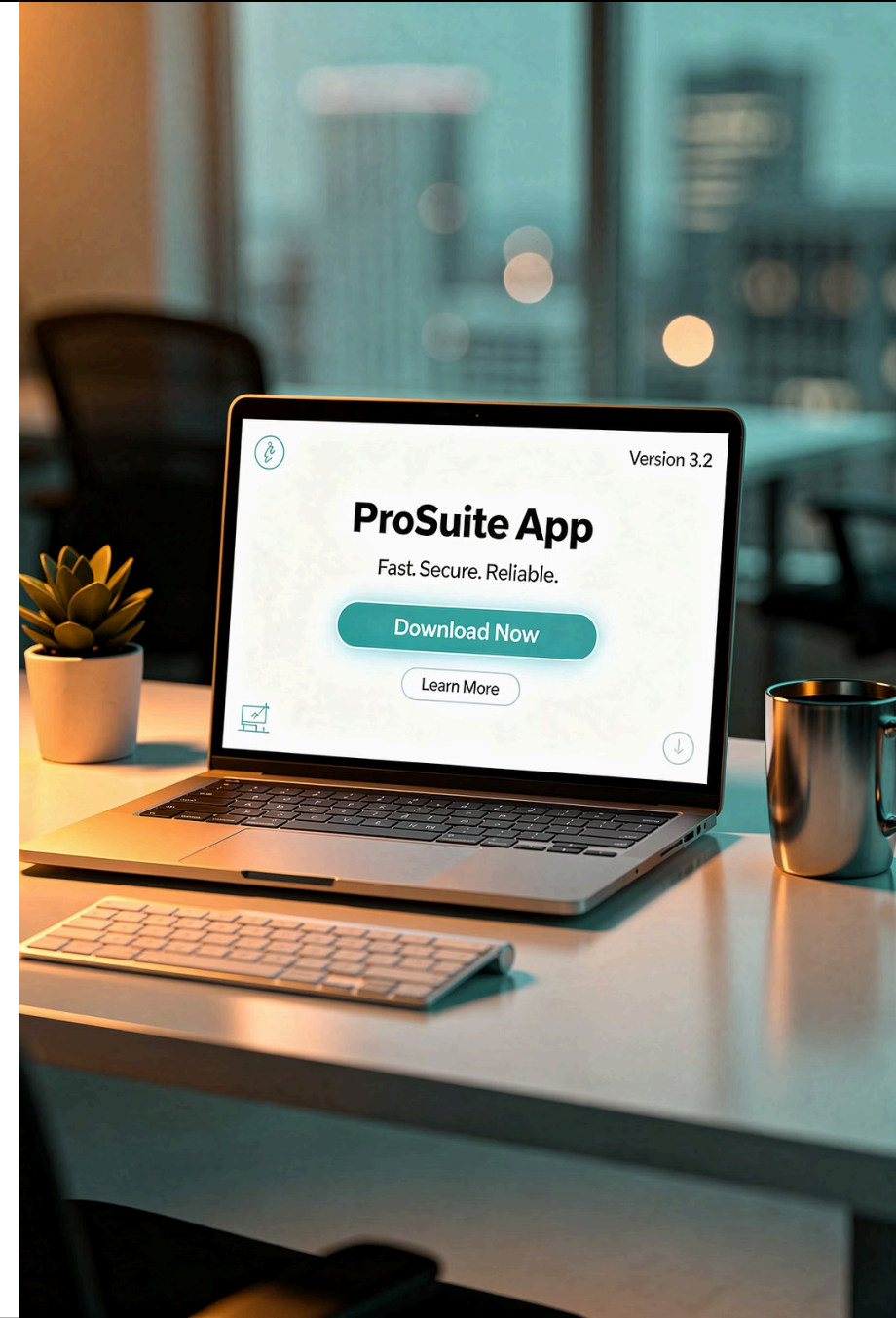
Primarily Windows and Linux – very stable, often used in professional environments.

## Hyper-V

Windows Pro/Enterprise/Education – integrated directly into Windows, good for Microsoft environments.

## KVM/QEMU

Primarily Linux – very powerful, more technically demanding.



# The right software for your system

## Windows Home

VirtualBox is often the easiest, as Hyper-V is usually not fully available.

## Windows Pro/Enterprise

Hyper-V or VirtualBox can be used. Note: Hyper-V can affect other solutions.

## Linux

VirtualBox or KVM/QEMU are common and well-supported.

## macOS

VirtualBox is possible; for newer Apple Silicon systems, additional restrictions apply—check the manufacturer's documentation.

# Installation: Typical Process

01

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## Visit the official manufacturer website

Always use the official source to ensure you receive secure and up-to-date versions.

02

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## Download the version for your host OS

Select the Windows, Linux, or macOS package.

03

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## Start the installation file & confirm default options

For beginners, the default options are usually sufficient.

04

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## Allow network drivers

A brief interruption of the network connection is normal – virtual network adapters are being set up.

05

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## Start the software empty

Start without a VM first to check if the installation is running correctly in principle.

# Hardware Requirements

A VM is "virtual", but it uses **real resources**. These requirements are critical:



## Hardware Virtualization

**Intel VT-x** or **AMD-V** must be supported by the processor and **enabled in the BIOS/UEFI**. Without this feature, some VMs will not start at all.



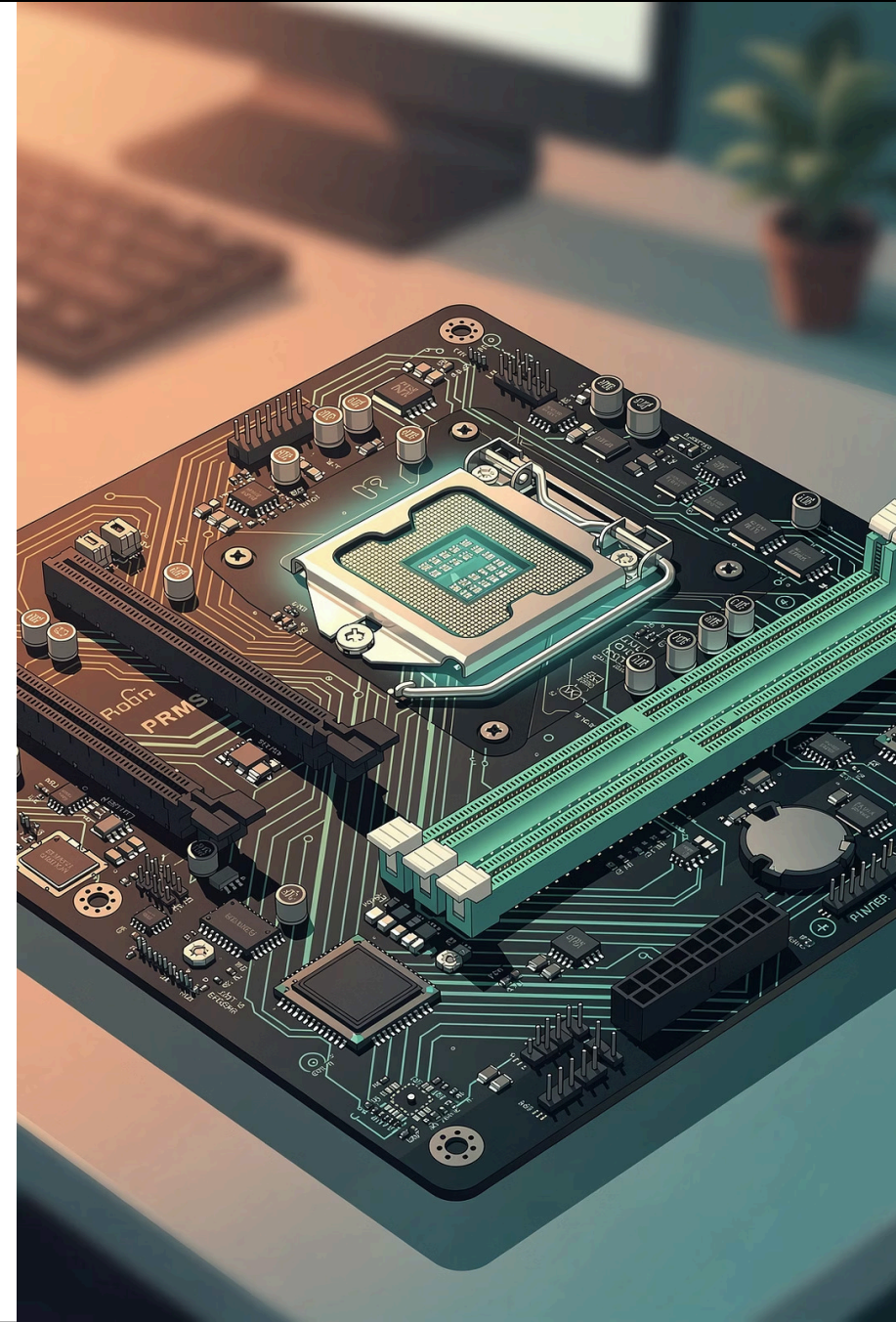
## Random Access Memory (RAM)

RAM is the most common bottleneck. The VM takes memory from the host – having too little will make **both** sides slow.



## Hard Drive & SSD

An **SSD** makes VMs significantly faster than an HDD. Always plan for extra storage space, especially for snapshots.



# RAM Guidelines for Virtualization

## 4 GB

**Total RAM**

Only useful for very small tests

## 8 GB

**Total RAM**

Simple Linux VMs often work  
fine

## 16 GB

**Total RAM**

Significantly more comfortable  
for learning purposes

## 32 GB+

**Total RAM**

Good for multiple VMs or  
heavier systems

Important: The host always needs its own reserves. Whatever is given to the VM is missing from the host system.

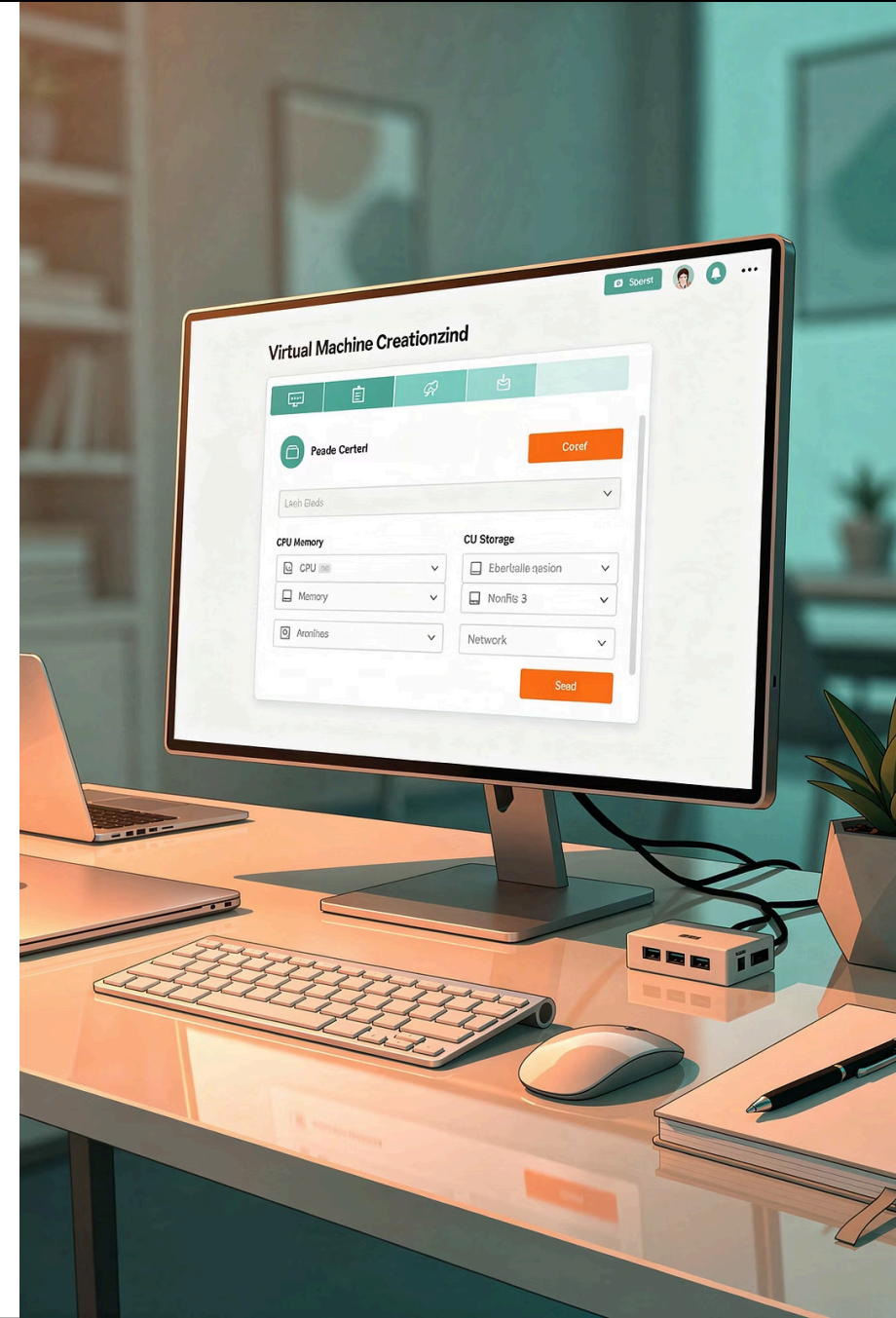
# Pre-flight Checklist

| Question                                | Why it matters                                   |
|-----------------------------------------|--------------------------------------------------|
| Does my CPU support virtualization?     | Otherwise, the VM may not run                    |
| Is virtualization enabled in BIOS/UEFI? | Common cause of startup errors                   |
| Do I have enough RAM?                   | Otherwise, the host and VM will become very slow |
| Do I have enough SSD/hard drive space?  | OS and updates require space                     |
| Are there conflicts with Hyper-V?       | Can interfere with VirtualBox/VMware             |
| Do I have an installation medium (ISO)? | No installation without an ISO file              |

## CHAPTER 2

# Creating and configuring virtual machines

A new VM is initially an empty container. Only through the correct configuration — name, RAM, CPU, hard drive, network — does it become a usable virtual computer.



# New VM Creation: Step by Step



The exact interface varies depending on the software, but the basic concept is almost the same everywhere.

# Name, Type, and ISO file

## Meaningful naming

Good names help to distinguish multiple VMs. Instead of *VM1* or *Test*, use:

- ubuntu-cli-basics
- windows-lab-update-test
- debian-network-learning

This way, you learn **clean technical work** at the same time.

## Type and ISO file

Select the correct guest operating system (Linux, Windows, BSD) and the version – the software will set appropriate default values accordingly.

The **ISO file** is an image of the installation medium. It is mounted as a virtual DVD drive. Upon the first startup, the VM boots from it and installs the guest operating system.

# Assigning CPU and RAM

## CPU: Enough, but not too much

For a simple lab VM with 8 logical host CPUs, **1–2 virtual CPUs** is reasonable. Too many CPUs do not make the VM faster, but put strain on the host. Scheduling becomes unnecessarily complex and the entire system reacts more sluggishly.

## RAM: Practical guidelines

**Small Linux VM without GUI:** 1–2 GB

**Linux with Desktop:** 2–4 GB

**Windows 10/11 for learning purposes:** 4–8 GB

Always plan in relation to the available host RAM.

- ❏ The golden rule: **Give the VM enough, but not too much.** The host always needs its own reserves – if you take too much from the guest, both will suffer in the end.

# Create Virtual Hard Disk

## Dynamically allocated

The file initially occupies little space and grows as data is written in the guest.

- Saves storage space initially
- Practical for learning environments
- Quick to set up

## Fixed size

The chosen storage space is reserved immediately.

- More predictable in operation
- Sometimes more performant
- Occupies a lot of storage space immediately

### Linux test system

20–30 GB

### Linux with desktop

30–50 GB

### Windows learning system

64 GB or more

# Understanding Network Modes

| Mode             | Meaning                                       | Typical Use Case                             |
|------------------|-----------------------------------------------|----------------------------------------------|
| NAT              | VM uses the host's network indirectly         | Simple internet access – ideal for beginners |
| Bridged          | VM appears as its own computer on the network | Testing in a real LAN, server services       |
| Host-Only        | Connection only between host and VM           | Isolated, controlled learning environment    |
| Internal Network | Connection only between VMs                   | Laboratory networks                          |

# Recommended Initial Configuration for Beginners



## CPU

1–2 virtual CPUs



## RAM

2–4 GB (Linux)  
4–8 GB (Windows)



## Hard Drive

Dynamic, 20–50 GB depending on guest



## Network

NAT – simple and secure for initial tests

This baseline is stable, simple, and easy to manage – ideal for learning.

# Adjusting Basic VM Settings

## → Boot Order

Set ISO/DVD first for installation. After a successful installation, move the **virtual hard disk** to the first position so the VM does not restart the installer.

## → Graphics and Display

Video memory, 2D/3D acceleration, and resolution affect desktop guests. For simple learning systems, conservative default values are sufficient.

## → EFI/UEFI or Legacy BIOS

Modern guest operating systems often use UEFI. If you encounter startup issues, it is worth checking this setting.

## → Guest Additions / VMware Tools

Significantly improve mouse integration, resolution, clipboard sharing, and folder sharing. These should be installed after the operating system installation.

# Snapshots: Essential for Learning

## What is a snapshot?

A snapshot saves the **state of the VM at a specific point in time**. You can return to it at any time—ideal for experimenting without risk.

## When to use snapshots?

- Before testing new software
- Before configuration changes
- When intentionally triggering errors
- As a recovery point before updates

**Important:** Snapshots are not a complete substitute for backups. They also significantly increase storage consumption.

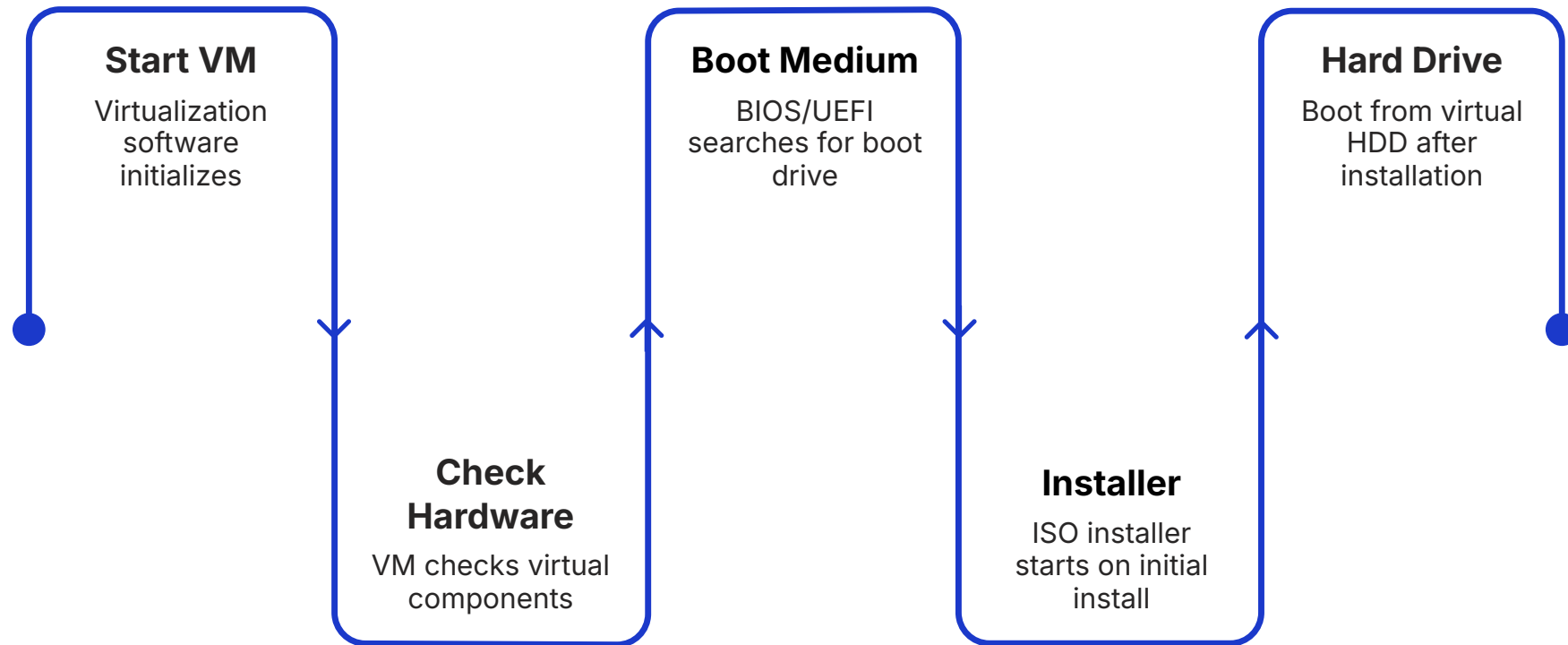


## CHAPTER 3

# First steps with the virtual machine

Once the VM is configured, the practical part begins: installing the guest operating system, performing initial steps, and observing the interaction between hardware, network, and the operating system.

# Typical VM Boot Sequence



A VM is not just a "program window with Linux or Windows," but a **virtual computer with a real boot process** – this is an important learning point.

# First Tasks After Installation

## 1 Check System

Check normal boot, language, time zone, and network connection.

## 2 Run Updates

Bring the operating system up to date.

## 3 Install Guest Additions

Improve mouse integration, resolution, and clipboard.

## 4 Monitor

Keep an eye on CPU usage, disk space, network access, and response time.

- ❏ The **Host Key** (right Ctrl key by default in VirtualBox) releases the mouse and keyboard back to the host computer if the VM has "captured" them.



## CHAPTER 4

# Identifying Configuration and Startup Issues

Problems at the beginning are normal. It is crucial to identify **systematically** at which level the issue lies: host, VM configuration, or guest operating system.

# Most Common Issues and Their Causes

| Symptom                              | Level            | Typical Cause                                       |
|--------------------------------------|------------------|-----------------------------------------------------|
| Error message regarding VT-x/AMD-V   | Host             | Virtualization disabled or blocked in BIOS          |
| Black screen immediately after start | VM Configuration | No bootable medium found                            |
| Installer restarts repeatedly        | VM Configuration | ISO still mounted, hard drive not at first position |
| No Internet in the VM                | VM/Guest         | Incorrect network mode or guest configuration       |
| VM is very slow                      | Host + VM        | Insufficient RAM, slow HDD, missing Guest Additions |
| Poor resolution / mouse issues       | Guest            | Guest Additions/Tools missing                       |

# Common Issues in Detail

1

## VM will not start

Hardware virtualization disabled in BIOS/UEFI, Hyper-V conflict, or insufficient free RAM. References to **VT-x**, **AMD-V** or **hypervisor** in the error message point to the host level.

2

## VM extremely slow

Insufficient RAM, VM is on HDD instead of SSD, host overloaded, Guest Additions missing, or updates running in the background. Slowness is rarely mysterious – usually a resource or driver issue.

3

## Storage full

**Distinguish between both levels:** Is the virtual disk file on the host full, or does the guest operating system itself have no space left? A typical beginner mistake is to only check the VM.

# Systematic Troubleshooting: The Right Order



## 1. Check Host

Is the virtualization software running normally? Enough RAM, CPU, storage? Virtualization enabled? Hyper-V conflicts?



## 2. Check VM Configuration

CPU, RAM, virtual hard disk, ISO, boot order, network mode – is everything set correctly?



## 3. Check Guest Operating System

Correctly installed? Driver issues? File system corrupted? Network service active? Updates failed?

This mindset – **separating layers cleanly from one another** – is the real learning gain and distinguishes structured technical work from mere trial and error.

# Conclusion: What you really learn

## Not just getting a VM up and running

But understanding **why** it runs – or why it doesn't.

## Thinking in layers

Cleanly separating host, VM configuration, and guest operating system – that is core tech competence.

## Working cleanly

Good naming conventions, conscious resource allocation, and snapshots make you faster and more confident.

